

Nathaniel Tunggal

PSCI 222

Stuxnet and the Theory of the Cyber Revolution

Over the past few decades, technology has become a core foundation of modern life, transforming the way societies work, communicate, learn, and experience the world. Thus, the potential for the networks and systems that underpin our lives to be attacked and weaponized poses an increasingly serious threat. This threat is the basis of the theory of the cyber revolution, which argues that cyberweapons have radically transformed how international politics fundamentally works. Stuxnet, the most sophisticated cyberattack to date, is often taken as proof of the theory of the cyber revolution; however, when one analyzes the roles of asymmetry, the offense-defense balance, and deterrence in the Stuxnet attack, it is clear that the skeptics make a stronger case against the theory of the cyber revolution. The Stuxnet story does not suggest that we are now living in a fundamentally different international political system in the cyber age.

We must first understand the concerns of the theory of the cyber revolution and arguments regarding how Stuxnet validates these concerns. The theory of the cyber revolution has three main pillars: cyberweapons asymmetrically benefit weaker actors while disadvantaging strong actors; cyberweapons are offense-dominant; and deterrence is largely ineffective against cyberweapons.

It is often argued that cyberweapons asymmetrically benefit weaker actors while disadvantaging stronger actors. Not only has creating and accessing cyberweapons become easier than ever, but so has attacking powerful nations due to their overwhelming reliance on networks and digital-dependent infrastructure. Unlike physical weapons, many potent hacking tools can be bought online or stolen from more powerful adversaries, empowering “a variety of nontraditional

actors, such as religious extremist groups, political activists, criminal syndicates, and individuals” (Kello, 36). Stuxnet is no exception, as “held against a single \$2 billion B-2 bomber, it was a Costco bargain” (Perlroth, 122). If such a sophisticated weapon like Stuxnet were to “leak or be sold or be given up to a group that wants to change the world order, we’re incredibly vulnerable” (Kello, 31). The result is that even if the cost of creating weapons are high, there are still ways in which weaker actors may obtain and use cyberweapons. On the other hand, because developed nations such as the United States have many developed networks and infrastructure, these networks are easier to penetrate due to a more complex defense surface. Because such large systems are so complicated and often rely on third-party equipment, there are more vectors of entry for attackers and more vulnerabilities to exploit. Cyberweapons by nature increase the potential capabilities of weaker actors while decreasing the defense ability of stronger actors.

Offense dominance is another key component of the theory of the cyber revolution. There are many reasons that underlie this claim. One is that attackers often take advantage of “zero-day vulnerabilities,” which, by definition, are impossible to defend against. Malicious code in infrastructure can also be incredibly difficult to detect. Stuxnet utilized seven zero-days, and it took several years for the Iranians to realize that they were under attack. Moreover, the psychological confusion hindered the defense and delayed discovery of the worm. In more extreme cases, an attack could go completely unnoticed due to such well-hidden malware. Defense fragmentation and supply-chain risks additionally disadvantage defenders (Kello, 29). Defense is further complicated by a general preference for convenience over security and a lack of a culture of security. Along with the comment that “what was considered a sophisticated cyber attack only a year ago... might be incorporated into a downloadable and easy to deploy internet application requiring little or no expertise to use” (Kello, 30), defense is not only extremely

costly, but inconvenient and slow-developing as well. As Lindsay puts it, “the defender has to succeed every time against every attack, but the attacker only has to succeed once” (Lindsay, 376).

That deterrence is largely ineffective in the cyber domain is a third key component of the theory of the cyber revolution. Joseph Nye names four mechanisms that “reduce and prevent adverse actions in cyberspace: threat of punishment, denial by defense, entanglement, and normative taboos” (Nye, 54-55). Because of the widespread proliferation of cyberweapons, in combination with the fact that “cyberspace affords an attacker an inordinate degree of anonymity” (Kello, 33), attribution becomes difficult. Had Stuxnet not been leaked, it would have been unlikely that the attack would be attributed to the United States and Israel due to the nearly perfectly hidden nature of the attack. If such attacks are possible, then in theory an inability to attribute the attack would eliminate the threat of punishment. Offensive dominance greatly reduces the chances of denial by defense, while blowback effects from entanglement can be limited through careful tuning of the attack, as seen by Stuxnet. Finally, a lack of a substantial history of cyber attacks creates a “lack of agreed standards of proportionality” (Kello, 35), obscuring normative taboos for a potentially massive cyber attack. In summary, a well-developed cyberweapon is able to be covert and anonymous, greatly reducing the victims’ ability to attribute attacks to attackers and reducing the role of traditional deterrence mechanisms in the cyber domain.

Stuxnet, however, was not a weapon weaker actors could have created. Even if weaker actors obtained a similarly powerful cyberweapon, they would still need the skill and knowledge to effectively utilize the weapon. Stuxnet, in order to covertly destroy the Natanz centrifuges, had to be designed exactly for the particular systems and centrifuges used in Natanz. Not only did the

Stuxnet creators need to “master the ICS configuration including its network architecture, its particular peripherals, the enrichment process it controlled, the organizational management of the plant, and the ways in which the system was likely to degrade upon malfunction” (Lindsay, 385), but “to engineer the ICS payload, developers would need access to IR-1 centrifuges, SIMATIC software, and the peripherals install at Natanz” (Lindsay, 386). The creation of Stuxnet required years of meticulous planning, organization, and intelligence assets. A less advanced group would be totally incapable of using or producing a similar weapon. Additionally, strong states, due to their wealth of resources, are both able to “manage the strategic risks of failure” (Lindsay, 388) in using cyberweapons and retaliate against weaker actors that do succeed in attacking the strong, creating a deterrence for cyberweapon usage by weaker actors. Thus, cyberweapons only extend the power of strong states, effectively keeping the barrier of entry for cyber weapon usage high by deterring usage by weaker actors. These arguments effectively refute cyber revolution supporters’ concern over the alleged asymmetry of cyberweapons.

Stuxnet also fails to provide compelling evidence for the claim that cyberweapons are offense-dominant. Stuxnet had to act completely undetected. Therefore, the attackers designed the worm to be more conservative to avoid detection, as “too much and Iranians would know they were under attack” (Lindsay, 392). Because the attack had to navigate a complex defensive surface, the potential for unknown failures hindered offensive power by deterring the creators from doing more than absolutely necessary. In addition, lower cost, open-source security likely utilized by Natanz could have actually increased offensive costs, as more work was necessary to break Iran’s defenses. While it is true that Stuxnet’s code “could also take out centrifuges not just in known facilities but in unknown ones” (Zetter, 197), it was important to specifically tailor the attack to Natanz due to concerns over being discovered, collateral damage, and entanglement

effects. The offense may have power to break into target systems, but at the same time it is severely constrained by a complex defense, high costs, uncertainty, and risks of collateral damage.

Finally, Stuxnet shows that deterrence is still effective. After all, the decision to use a cyberattack on Iran's nuclear facilities only arose from successful deterrence against using conventional weapons, and Stuxnet's limited offensive capabilities were mainly due to deterrence from being discovered. While these instances of effective deterrence are obvious, supporters of the cyber revolution argue that because attribution in the cyber domain is more difficult, the threat of punishment becomes ineffective. However, detection and attribution are rarely impossible, leaving deterrence still effective. A non-zero chance of being discovered can be enough to deter an attacker; if discovered, states have the resources to pursue those who do significant damage. This is analogous to how, on a smaller scale, there is "plenty of spam for fake pharmaceuticals, but no spam pushing Schedule II drugs. Criminals are deterred from energizing a more aggressive law enforcement response" (Lindsay, 401). Even when attribution takes a long time, as in the case of Stuxnet, attribution time is still an unknown that attackers cannot take for granted. Even an attack as sophisticated as Stuxnet was able to be attributed to the US and Israel. Thus, attackers cannot rely on anonymity.

The cyber domain is still a relatively new space, and there is no doubt that there are many unknowns regarding the role of cyberspace in international relations. Arguments supporting the theory of the cyber revolution largely rely on logical and ideal-type situations. While these arguments do hold value, they fall short. At the same time, Stuxnet is only one example of a major cyberattack—analyses and conclusions drawn from Stuxnet alone are not enough to clearly define the role of cyberweapons in the international political system. That being said,

because Stuxnet conformed to a more traditional political system—it did not asymmetrically increase the power of weaker actors, its offense was mitigated by a strong defense, and it played by traditional rules of deterrence—skeptics make a more persuasive claim against the theory of the cyber revolution. Even though a cyber revolution may not be on the horizon, the cyber domain continues to play an increasingly important role in international politics, so it is crucial to better understand the unknowns in order to get ahead of the cyber evolution.

Works Cited

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